

MarketMind AI - Milestone 2 Concise Theory

A compact explanation of the objective, datasets, models, integration, role access and current status.

Objective

Milestone 2 adds customer intelligence and machine-learning forecasting to MarketMind. Cleaned business data is converted into predictions, stored in the database and delivered through secured FastAPI endpoints to role-specific React dashboards.

1. Core intelligence features

Customer segmentation

The Online Retail II dataset is used to calculate Recency, Frequency, Monetary value, engagement and return behaviour. K-Means is the selected unsupervised model, with Hierarchical clustering used for comparison. It groups 5,878 customers into Champions, Needs Attention and Hibernating. The current silhouette score is 0.364, so the segments are useful for initial decisions but still require business review.

Business revenue forecast

Amazon sales records are aggregated into daily net revenue in INR. A weekday-aware Linear Trend model currently gives the lowest error among the evaluated candidates. It provides the Business Owner with 7, 14 and 30-day revenue predictions, confidence bounds, growth, MAE and RMSE.

Personal sales forecast

Processed sales transactions are filtered to the signed-in Sales Executive. Seasonal Naive is used because the seller history is limited and recent weekly patterns provide a stable baseline. The response contains historical sales, predictions, growth, trend and uncertainty, without exposing another seller's data.

Product-demand forecast

Train and evaluation Parquet data provides product, store, time and demand features. XGBoost Regressor is selected for its ability to learn non-linear relationships. Current evaluation is approximately MAE 2.152 and R2 0.918. The 7, 14 and 30-day predictions are compared with inventory to provide demand trends and stock-risk context for the assigned Store Manager.

Integration, APIs and Access Control

The backend is the integration and security boundary; the frontend never reads the database directly.

2. End-to-end workflow

Connected workflow

1. Raw datasets are cleaned and validated.
2. Features are generated and model candidates are compared.
3. Predictions and model metadata are saved as versioned outputs.
4. Duplicate-safe import commands insert or update database records.
5. FastAPI verifies the token, session, MFA, role, permission and tenant/store/seller scope.
6. SQLAlchemy reads the allowed records and React renders the KPIs, charts, filters and tables.

3. Main API contracts

Endpoint	Purpose	Allowed access
POST /api/v1/auth/login	Access and refresh tokens	All roles; Admin supplies MFA
GET /dashboard/access	Permitted modules/actions	Authenticated user
GET /customer-segments/summary	Segments and behaviour KPIs	Authorised business/store/seller scope
GET /forecasts/revenue	Revenue series and metrics	Owner and Administrator
GET /forecasts/personal	Personal sales forecast	Executive; Admin-selected seller
GET /forecasts/demand	Demand and stock-risk context	Manager; Admin-selected store
GET /forecasts/monitoring	Models, health and jobs	Administrator after MFA

4. Role-based access

Role	Milestone 2 access	Restriction
Business Owner	Business segments and revenue forecast	No model administration
Store Manager	Assigned-store summary and product demand	No business-wide or other-store data
Sales Executive	Assigned customers and own forecast	No other seller's records
Administrator	All forecast types and monitoring after MFA	Selectors remain tenant-bound

Security principle

Frontend visibility is not the security control. Every protected request is checked again by FastAPI. Invalid input returns 422, missing data returns 404 and insufficient permission returns 403 Forbidden.

Testing, Status and Honest Limitations

What is verified now and what should be explained clearly during evaluation.

5. Verification

Area	Verified result	Coverage
Backend	17 tests and Ruff passed	Authentication, MFA, RBAC, scopes, APIs and imports
Preprocessing / ML	10 tests passed	Cleaning, features, clustering, candidates and horizons
Database	Alembic check passed	Models and migrations remain aligned
Frontend	Oxlint and production build passed	Real API UI compiles successfully

6. Current status

Functionally integrated for local evaluation

The datasets, preprocessing pipelines, selected ML models, database records, secured FastAPI APIs and React dashboards are connected. The four roles receive different views and backend-enforced data scopes. Import operations are duplicate-safe, and forecast filters and horizons are validated.

Limitations to state honestly

Revenue accuracy will improve with longer and more recent history. Product-demand and inventory data still need a fully confirmed shared product/store mapping, and the Parquet sale_amount meaning must stay clearly documented. Customer clusters need periodic business review. Drift monitoring, formal model governance and production deployment controls are future hardening work.

7. Final explanation

One connected Milestone 2 system

MarketMind is not only an ML notebook or a frontend mock-up. Data is cleaned, models are compared, results are versioned and imported, FastAPI applies backend security, and React displays only the information permitted for the logged-in role. The Owner receives revenue intelligence, the Manager receives demand and stock-risk information, the Executive receives a personal forecast, and the Administrator monitors the forecasting system after MFA.